

- A. Requirement Set A - Identification and Preparation
 - a. Req. A. 1: The design shall automatically sort Robertson screws into distinct vessels based on each variant
 - b. Req. A. 2: The design shall classify Robertson screw lengths within the range $\frac{3}{4}$ to 3 inches at $\frac{1}{4}$ inch intervals with $\geq 97.5\%$ accuracy.
 - i. test: Check if after sorting a batch of screws over 97.5% of them are in the correct length container
 - c. Req. A. 3: The design shall classify Robertson screw by different head shapes as defined in Appendix B with $\geq 97.5\%$ accuracy.
 - i. test : Check if after sorting a batch of screws over 97.5% of them are in the correct head shape section
- B. Requirement Set B - Efficiency and Automation
 - a. Req. B. 1: Processing time per individual screw shall be ≤ 5 seconds
 - i. test: the time to sort a batch of N number of screws should be less or equal to $5*N$, therefore in a test with N number of screws the time from inserting the screws into the machine to retrieving the containers should be less than $N*5$.
 - b. Req. B. 2: The design shall operate continuously without user interaction after putting in the screw
 - i. test: Check that the device works as intended without user interaction apart from inserting the screws
- C. Requirement Set C - Usability and Training
 - a. Req. D. 1: The system shall fit within a 30*30*30 cm bounding volume during all phases of operation
 - i. test : Measure the device
 - 1. measure the device
 - b. Req. D. 2: The total dry mass of the device (unloaded) shall be $\leq 4.5\text{kg}$
 - i. test : weigh the device
 - 1. weight the device
 - c. Req. D. 3: If electrically powered, the system shall operate using standard 120 V AC with a maximum current draw of 15 A
 - i. test (power versatility): also calculations can be done
 - d. Req. D. 4: The system shall maintain operational performance while exposed to airborne sawdust for ≥ 100 screws sorted
 - i. test (durability) : Test 100 screws sorted to confirm the device keeps working as intended after this

Evaluation criteria

the design is better when

1. Accuracy of classification for screw length and head shape (the higher the accuracy, the better)
2. throughput efficiency as reflected in the processing time per individual screw (the less time per screw, the better)
3. Autonomous capacity as measured by the number of screws the device can process in a single cycle without user intervention (the more screws, the better)
4. Compactness of the system as reflected in the total bounding volume (the less volume, the better)
5. Mass efficiency and portability of the design (the less mass, the better)
6. Environmental robustness and the ability to maintain performance while exposed to airborne sawdust (the more operational cycles, the better)
7. Clarity of operational feedback as reflected in the visibility distance of status indicators (the greater the distance, the better)

8. Information density of the instructions, providing full operational knowledge in minimal space (the shorter the text, the better)

EC1 and EC2 are assigned the highest weights (justification: the stakeholder interviews and the primary need for reliable, high-speed automated sorting)

testing method

1. Requirement: A.2:
 - a. The requirement states that lengths shall be sorted in the specified lengths categories with an accuracy of $\geq 97.5\%$ accuracy
 - b. The test will be performed twice, once on the length sorting mechanism alone and once on the final prototype before testing.
 - c. In both cases 3 consecutive batches of 20 known length screws will be used, in both cases the accuracy will be an average of the 3 batches.
 - d. For the first test the screws will be placed manually into the mechanism, for the final test the screws will be placed automatically with the mechanism.
 - e. If the mechanism passes the first test it will be deemed acceptable and we will move forward with that design for the final prototype, if on the other hand the design fails the first test, modifications will be made to make sure the final prototype meets the requirement
2. Requirement: A.3:
 - a. The requirement states that head shapes shall be sorted in the specified head shape categories with an accuracy of $\geq 97.5\%$ accuracy
 - b. The test will be performed three times, once to test raspberry pi + camera + code integration, once with the proposed stand + lighting environment and one last time in the final prototype for showcase.
 - c. For all 3 tests 3 batches of 20 screws will be analysed, this makes a total of 60 screws. The accuracy will be an average of the 3 batches.
 - d. For the first test a "perfect" test environment will be used, meaning the lighting and background contrast will be perfect for the camera to detect the headshape, for the second test the screws will be placed manually into the test stand while for the final test the screws will enter the head shape analysis section automatically.
 - e. If the mechanism passes the first test the code + camera + raspberry pi integration will be deemed acceptable, if the mechanism passes the second test the final mechanism will be deemed acceptable for the final prototype.